Automated Test Report Generation System

Challenge Overview

Testing laboratories generate a large number of test reports every year. Data required for these reports is collected from multiple sources such as customer request forms, test logs, measurement records, and laboratory databases.

Currently, preparing the final test report often involves manually collecting information, validating entries, and populating report templates. This process can be time-consuming, prone to human errors, and may lead to delays in delivering reports to customers. The challenge is to develop a solution that automates report creation, improves accuracy, and significantly reduces report preparation time.

Problem Statement

Design and develop an automated test report generation system that can:

- Collect data from multiple sources.
- Store and manage information in a database.
- Automatically populate a predefined report template.
- Generate complete test reports with minimal manual intervention.
- Reduce report preparation time and improve accuracy.

The solution should demonstrate how data from customer requests, test records, and laboratory log sheets can be integrated into a single automated workflow.

Objectives

The proposed solution should aim to:

1. Reduce Report Generation Time

Automate repetitive tasks and speed up report preparation. [Hackathon_ATRG-SCL | PDF]

2. Minimize Manual Errors

Improve data consistency and accuracy through automated data processing.

3. Improve Productivity

Reduce effort spent on manual report creation and review.

4. Enhance Customer Experience

Enable faster delivery of reports and better service quality.

Expected Solution

Participants are expected to develop a web-based software prototype that includes:

Data Collection Module

- Import customer request information
- Import laboratory log sheets
- Store test results in a database

Database Management Module

- Organize and manage test-related data
- Enable retrieval of historical records

Report Generation Engine

- Automatically populate report templates
- Generate reports in PDF or printable formats

Dashboard

- View report status
- Search test records
- Track report generation progress

Suggested Workflow

1. Capture customer request details.
2. Import test and measurement data.
3. Validate and store data in a database.
4. Populate the report template automatically.
5. Generate the final report.
6. Review and export the report.

Functional Requirements

The system should support:

Input Data Sources

- Customer request forms
- Laboratory log sheets
- Test measurements and results
- Existing databases or spreadsheets

Processing Features

- Data validation
- Template mapping
- Automatic report generation
- Database storage and retrieval

Output Features

- Formatted test reports
- Searchable report records
- Downloadable reports
- Summary dashboard

Expected Deliverables

Teams should submit:

Working Prototype

A functional software application demonstrating automated report generation.

Source Code

Well-documented code with installation and usage instructions.

Architecture Diagram

System architecture showing data flow and major modules.

Demonstration

A live demo or recorded walkthrough of the solution.

Technical Presentation

A short presentation explaining:

- Problem understanding
- Solution architecture
- Technologies used
- Results achieved